

CryptoChain Analyzer Dashboard

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This Streamlit dashboard connects to public Bitcoin APIs and displays live cryptographic metrics related to SHA-256, Proof of Work, block headers, Merkle roots and difficulty adjustment.

1. Cryptographic Metrics Displayed

M1 monitors Bitcoin mining. It fetches the latest block, displays hash, height, nonce, transaction count, compact bits and difficulty, converts bits into the 256-bit Proof of Work target, compares the block hash with that target and visualizes the leading-zero threshold.

It estimates network hash rate with: $\text{hash rate} = \text{difficulty} * 2^{32} / \text{observed average block time}$. It also plots recent inter-block times. Since mining attempts are independent trials, the expected waiting-time baseline is an exponential distribution.

M2 parses the real 80-byte Bitcoin header and shows version, previous block hash, Merkle root, timestamp, bits and nonce. The code uses hashlib to compute $\text{SHA256}(\text{SHA256}(\text{header}))$, reverses byte order for explorer format, checks the API hash, validates the target and counts leading zero bits.

M3 analyzes completed 2016-block difficulty periods. It marks adjustment events and computes the ratio between actual elapsed time and the expected $2016 * 600$ seconds.

2. AI Model: Anomaly Detector

For M4, I chose the anomaly detector option. The model trains on recent Bitcoin block inter-arrival times from live Blockstream data. It uses an exponential distribution because Bitcoin mining is commonly modeled as a Poisson process.

The detector fits the mean from the selected sample and flags extreme-tail intervals as Too fast or Too slow. Metrics shown are fitted mean, MAE versus 600 seconds, anomaly rate and total log-likelihood.

Model element	Implementation
Training data	Recent live Bitcoin block timestamps from Blockstream.
Feature	Seconds between consecutive blocks.
Baseline	Exponential distribution fitted from the selected sample.
Evaluation metrics	Fitted mean, MAE vs 600s, anomaly rate and log-likelihood.

Limitations: this is unsupervised, so it highlights statistically unusual intervals but does not prove malicious behavior.

3. Optional Modules

M5 verifies Merkle proofs by downloading txids, building a transaction branch, duplicating odd nodes when needed, recomputing every parent with double SHA-256 and comparing the final root with the real header Merkle root.

M6 estimates hash rate from live difficulty and block timing, then estimates energy cost, hardware cost and double-spend success probability versus confirmation depth using Nakamoto section 11.

M7 is a second AI approach: a difficulty predictor. It trains a transparent linear regression model on recent adjustment periods and evaluates it with walk-forward MAE and MAPE.

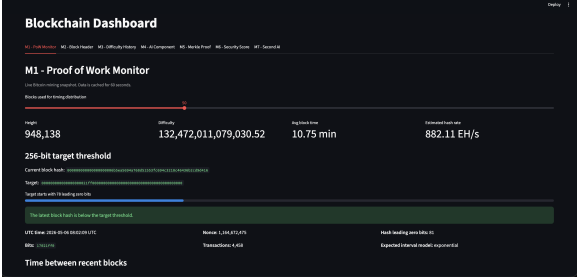
4. Dashboard Functionality

The app has seven Streamlit tabs, handles API errors with visible messages, caches live API calls for responsiveness and includes a 60-second browser refresh mechanism. The repository also includes screenshots for M1-M7 under images/.

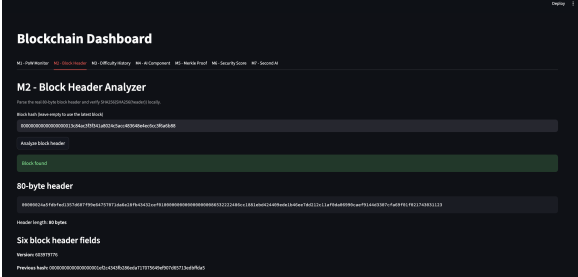
5. References

- Satoshi Nakamoto, Bitcoin: A Peer-to-Peer Electronic Cash System, 2008.
- Blockstream API documentation: <https://blockstream.info/api/>
- Blockchain.com charts API documentation: https://www.blockchain.com/explorer/api/charts_api
- Bitcoin Developer Reference: block headers and compact target representation.

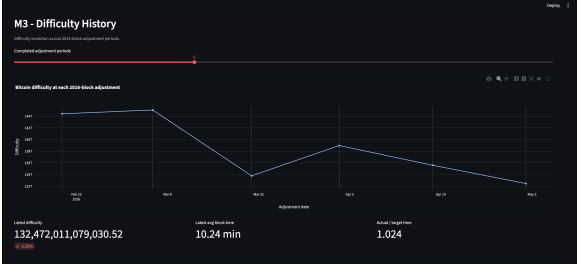
Screenshot Appendix



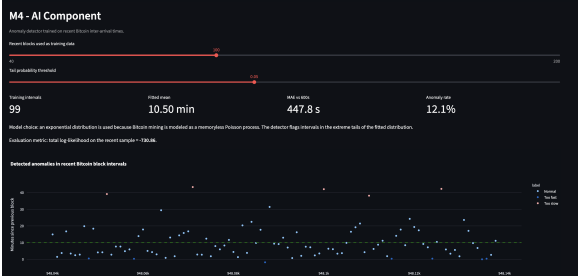
M1 - Proof of Work Monitor



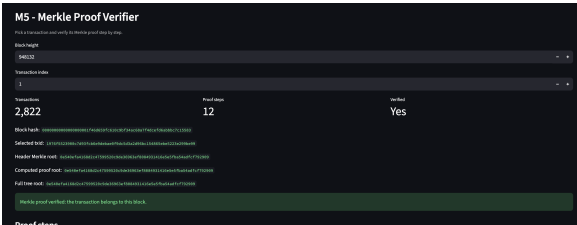
M2 - Block Header Analyzer



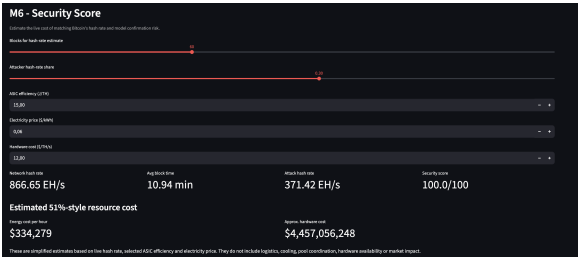
M3 - Difficulty History



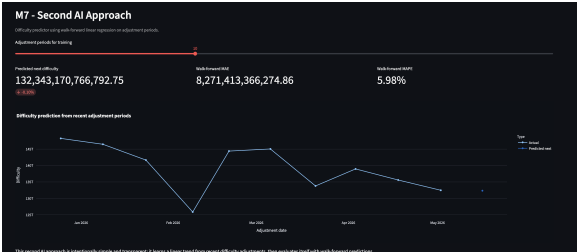
M4 - Anomaly Detection



M5 - Merkle Proof Verification



M6 - Security Score



M7 - Second AI Predicto